

Research Article

Analyzing and Detecting Abnormal Behaviors of Drug Abuse and Addiction Users in School Environments Based on Deep Learning Approaches

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Drug abuse and addiction problems are one of the most serious health, social, and psychological problems facing the world. Many international studies indicate that the start of drug abuse occurs mostly in adolescence, which is the period that young people spend in schools, institutes, and universities. Drugs in the student community have become a scourge that raises increasing concern, whether among families or educators, over the fate of school children and educational attainment. Regarding their behaviors, an addicted student often exhibits abnormal behaviors such as permanent lethargy, anxiety, tremors, and aggressive behavior toward others. Moreover, to obtain drugs, the addicted student becomes compelled to resort to various means and ways, and they gradually become criminal addicts. To this endeavor, a detector of abnormal behaviors in schools has become a necessity. In this paper, we built an automatic system able to analyze and detect abnormal behaviors of addicted students and inform the educational staff and parents to know how to manage and treat them. On a technical level, we used deep learning and the recent computer vision techniques in the suggested solution due to their contributions to human behavior and emotion recognition fields. The best-recorded result (97.5%) is obtained with fused handcrafted features based on skeleton joints and deep features extracted with the MobileNet pretrained model and forwarded to a deep proposed network based on two TimeDistributed layers, one BiLSTM layer, and several Dense layers.

Keywords: abnormal behavior recognition; deep learning models; drug abuse and addiction; pretrained CNN models

1. Introduction

Drugs are compounds that have an effect on the body and the brain. Different medications can have various impacts. Some pharmacological effects include long-term and irreversible health repercussions. They can even remain after the user has stopped using drugs.

Drugs can be consumed in a variety of ways, including injection, inhalation, and ingestion. The way medications are given affects how they affect your body. They abused narcotics, which had an impact on their brain. They flood the brain with

dopamine, a chemical that helps control our emotions, motivation, and pleasure experiences, resulting in a “high.”

In addition, drugs may ultimately affect how the brain functions and impair a person’s capacity to make decisions, resulting in extreme cravings and compulsive drug usage. This habit can develop into substance dependency, or drug addiction, over time.

Today, one in four fatalities is related to the use of illegal drugs, and more than 7 million individuals suffer from an illicit drug condition [1, 2]. In actuality, drug misuse is more closely linked to diseases, impairments, and fatalities than

any other sickness that can be avoided. Drug and alcohol addicts are more likely to sustain unintended injuries, be involved in accidents, and experience domestic violence [3].

Many international studies indicate that the start of drug abuse occurs mostly in adolescence, which is the period that young people spend in schools, institutes, and universities [4–7]. Moreover, drugs in the student community have become a scourge that raises increasing concern, whether among families or educators, over the fate of school children and educational attainment. Regarding their behaviors, an addicted student often exhibits abnormal behaviors such as permanent lethargy, anxiety, tremors, and aggressive behavior toward others. Moreover, to obtain drugs, the addicted student becomes compelled to resort to various means and ways, and they gradually become criminal addicts [2].

In this endeavor, a detector of abnormal behaviors in schools has become a necessity. In this research, we seek to build an automatic system able to analyze and detect abnormal behaviors of addicted students and inform the educational staff and parents on how to manage and treat them. While previous studies focused on individual feature extraction methods, our approach combines handcrafted features based on skeleton joints with deep learning models, specifically using transfer learning models MobileNetV2. This fusion enhances the accuracy and robustness of behavior detection, as demonstrated in our results.

This study's contributions are summarized as follows:

- **Dataset Preprocessing:** Techniques were used to eliminate noise and clean the sequences. Each sequence was modified to represent only one person while recording detailed movement information such as exact body position and gestures. This phase was critical for improving the clarity and dependability of the following feature extraction methods.
- **Handcrafted Feature Extraction:** We used the MediaPipe API, a platform designed for developing perception pipelines [8], to extract custom features based on skeleton joint [9] recognition. These joints are essential points in the human body that are recognized in video frames to evaluate human movement. Examples include elbows, knees, and shoulders. This method gave detailed spatial information about body movements, which is essential for correctly identifying abnormal behaviors.
- **Deep Feature Extraction Using Transfer Learning:** We used multiple transfer learning models [10], which are techniques where a model developed for one task is reused as the starting point for a model on a second task, to extract deep features from convolutional neural network (CNN) layers, which are designed to process data that have a grid-like topology. Specifically, we used pretrained networks like VGG19, Xception, ResNet100, and InceptionResNetV2, which are well-established models in the field of computer vision. These pretrained models are employed to collect high-level features that indicate both normal and abnormal student behaviors. These traits were then combined with the handcrafted ones to improve model performance.

- **Predictive Modeling Using CNN-BiLSTM:** Using a CNN-BiLSTM framework, a very accurate prediction model was created. This design was chosen because it can capture both geographical and temporal correlations in data, making it ideal for categorizing activities into two classes: normal (0) and abnormal (1).
- **Comprehensive Evaluation:** An empirical investigation was carried out to determine the best-performing classifier. The Dense layer, with a sigmoid activation function, was used in conjunction with many supervised learning approaches, such as support vector machine (SVM), random forest (RF), Naive Bayes (NB), and K-nearest neighbors (KNNs). This evaluation was crucial in finding the most effective classifier configuration for the proposed system.

The remainder of this paper is organized as follows: Section 2 conducts a review of the literature on human activity recognition (HAR). In Section 3, we present theoretical research that describes the impact of drugs and alcohol on student behaviors. Then, in Section 4, we present the proposed methodology toward abnormal behavior recognition systems related to drug and alcohols abuse by evaluating the fused handcrafted and the deep features. Section 5 presents the empirical study of our proposed system. Section 6 presents the quantitative and qualitative evaluations. Section 7 presents a discussion that demonstrates the effectiveness and precision of our suggested approach. Finally, Section 8 highlights the current work's important findings and indicates crucial directions for future works.

2. Related Works

HAR based on computer vision techniques involves analyzing video or image data to recognize and classify human activities. Here are some common computer vision techniques used in HAR: object detection, pose estimation, motion analysis, feature extraction, action recognition models, temporal modeling, and fusion techniques. So, in order to develop applications for HAR, several research works and studies used deep learning and computer vision techniques.

The authors of [11] described a method for detecting human activities in video sequences using local space–time characteristics and SVMs. The method detects localized motion events that are resistant to fluctuations in scale, frequency, and velocity and then utilizes these properties to identify six human movements (walking, jogging, running, boxing, hand waving, and hand clapping) from a dataset of 2391 sequences. Combining these characteristics with SVM classifiers results in excellent recognition accuracy across several contexts, exceeding previous approaches.

An action detection framework was proposed by Saha et al. [12] for the purpose of classifying and identifying the start and finish locations of actions in a video series. Two streams are used for frame-level detection and classification: RGB for appearance and optical flow for motion. Based on

spatial overlap and softmax probability values, these detections are merged. Next, using altered video frames, action cubes are constructed, gathering detections with spatial overlap and class-specific scores. To preserve the integrity of the label, action cubes are cut. Even with its excellent speed, the framework's nonincremental construction of action tubes with every new frame capture makes it unsuitable for real-time applications. Changes were proposed by Singh et al. [13] to enable early action prediction, while video frames were being streamed. The updated version has components that have been improved, but fundamentally it is comparable to [12]. To begin with, real-time optical flow reduces computation time for motion streams. For regression and classification, a single-shot multibox detector (SSD) is utilized. Third, action tubes are created frame by frame by an efficient yet greedy algorithm that recognizes actions based on previous frames. Action tubes are clipped after k consecutive frames of observation. This version has better accuracy and is seven times quicker. As the video progresses, performance in [13] is displayed, showing 48% accuracy after viewing the first 10% of the movies in the J-HMDB-21 dataset.

In addition, Carrara et al.'s lightweight architecture [14] annotates untrimmed human skeletal sequences in real-time using an LSTM-based network. In order to minimize latency while accurately identifying activity boundaries, frame-level annotation is employed. To guarantee precise action delineation, LSTM hidden states first give probability to each class for each incoming stream posture. These probabilities are subsequently adjusted by a learning threshold. The proposed method has a greater accuracy than online detection approaches and can annotate about 7700 frames in a second.

In [15], transfer learning is used in this work to extract human motion features from RGB video frames in order to increase detection accuracy. To extract descriptive features, a CNN based on the Visual Geometry Group network-19 (VGGNet-19) pretrained model is utilized. Then, to build a binary SVM model, the feature vector is fed into the binary support vector machine classifier (BSVM). Experiments performed on two different datasets with vastly different context abnormalities yielded an accuracy of 97.44% and an area under the curve (AUC) of 0.9795 for the University of Minnesota (UMN) dataset and an accuracy of 86.69% and an AUC of 0.7987 for the University of California, San Diego Pedestrian1 (UCSD-PED1) dataset. Furthermore, the results show that VGGNet-19 outperforms histograms of oriented gradients, background removal, and optical flow. Furthermore, VGGNet-19 outperforms other pretrained networks such as GoogleNet, ResNet50, AlexNet, and VGGNet-16 in terms of detection accuracy.

In [16], the authors developed an effective method for designing a system that identifies any unexpected or abnormal occurrence and informs the proper authorities. The entire surveillance system training approach may be divided into three stages: data preparation, model training, and inference. The framework is made up of two neural networks: CNN and a recurrent neural network (RNN). The CNN technique is used to extract high-level features from

frames in order to reduce the complexity of the input. The RNN is used for classification because it is well-suited for video stream processing. At this time, the model is being trained to predict behavior based on the film. The suggested approach achieved an accuracy rate of 95.3%.

Moreover, [17] introduced an effective crowd analysis mechanism based on a human behavior analysis model and a motion heat flow enabled optical flow approach. In this work, the object contact points (OCP) descriptor is used to recognize objects and build feature weights by evaluating needed object points from the motion heat map. In addition, by using the suggested human behavior analysis model, three types of histogram gradients are evaluated: discrepancies between adjacent points and OCP descriptors, gradients acquired considering spatial angle, and gradients obtained using temporal angle. Using a feature encoding approach, features gathered in both the spatial and temporal domains are encoded. A KNN classifier model is used to classify test data. The accuracy attained with the suggested HBA model is 99.64% for the violent-flow dataset and 99.24% for the web dataset, which is significantly higher than numerous existing human behavior monitoring approaches. Similarly, the accuracy and recall rates for the violent-flow dataset are 98.90% and 98.80%, respectively, while the precision and recall rates for the web dataset are 99.62% and 98.82%, respectively.

In [18], authors focused on identifying dangerous student actions automatically, which confronts several obstacles such as limited datasets, perplexing postures, keyframe identification, and rapid reaction. To solve these issues, Zhou et al. created the dangerous behavior dataset from surveillance films and then translated action recognition from lengthy movies into an object identification job that avoids keyframe detection. Then, they provided DangerDet, a unique end-to-end dangerous behavior detection technique that integrates multiscale body features with key points-based posture features. Because of the strong association between posture and behavior, DangerDet obtains 71.0% with roughly 11 FPS on our dataset. It achieved a better balance of accuracy and time cost.

In [19], the authors aim to identify and detect various abnormal events like violence and property destruction captured in CCTV recordings, which have a significant impact on public safety. In this study, the authors faced a main challenge such as the traditional anomaly detection methods, especially in real-time scenarios that require high computational resources. To address this challenge, Patwala et al. proposed a low-cost algorithm with high accuracy. This algorithm employed a CNN based on the DenseNet121 architecture. This CNN serves as a feature extractor to identify patterns and anomalies in the video data. The performance of the proposed algorithm is evaluated using the UCF-Crime dataset. The study reports an AUC of 86.63%, indicating its effectiveness in identifying abnormal events in the videos. The proposed algorithm achieves new state-of-the-art performance on the UCF-Crime dataset, suggesting that it outperforms existing methods in terms of accuracy and efficiency.

Many previous studies in the field of HAR and abnormal behavior detection, like those by Schuldts et al. [11] and Saha et al. [12], used SVM and optical flow-based techniques to focus on common human actions. However, because of their high computational complexity, these studies struggled to be applied in real-time and to scale. Pretrained models and LSTM networks were utilized in studies like Carrara et al. [14] and Al-Dhamari et al. [15]. These models demonstrated great accuracy but were frequently dataset-specific, raising questions about their generalizability in a variety of situations like schools. CNNs were successfully used by Zhou et al. [18] and Patwala et al. [19] to address dangerous activities in students and abnormal events; nevertheless, real-time performance and model complexity posed obstacles. The crowd analysis experts Pattan et al. [17] mostly relied on deep features, ignoring the possibility of constructed features that might offer crucial insights into more complex settings. Overall, these techniques performed well on some datasets, but they have drawbacks that need to be addressed for more effective and generalizable models, including the difficulty of training, the contextual applicability of the approaches, and the disregard for handcrafted features. Table 1 presents a comparative study of the mentioned related works.

3. Theoretical Study: Adverse Effects of Drugs on Body, Brain, and Behaviors of Addicted Students

Addiction is a chronic condition defined by obsessive substance seeking and use, notwithstanding the negative effects. Clinically, this is known as substance use disorder (SUD). Addiction theories, such as the illness model of addiction and behavioral psychology, argue that addiction is caused by complex neurological, psychological, and environmental processes rather than human decisions or moral failure [1]. According to the disease model, addiction changes brain function, notably in the brain's reward system (part of the limbic system), which is dominated by dopamine. This system regulates pleasure and reward, and drugs stimulate it by releasing excessive dopamine, resulting in extreme sensations of euphoria. Over time, the brain's chemistry adjusts, resulting in a dependence in which people require drugs to feel normal rather than to enjoy pleasure. From the perspective of behavioral psychology, the operant conditioning model explains addiction through reinforcement mechanisms. Drug use is positively reinforced by the immediate pleasurable advantages (the "high"), but negative reinforcement occurs when drug use alleviates withdrawal symptoms or mental distress. These methods may persuade people, particularly vulnerable teenagers, to continue taking drugs despite the negative consequences. Understanding the scientific origins of addiction explains why the following detrimental effects occur in student populations:

- Adverse effects of addiction student bodies [1, 20]:
 - A compromised immune system, which increases the risk of disease and infection.

- Heart issues induced by injectable drugs include abnormal heart rhythms, heart attacks, collapsed veins, and blood vessel infections.
- Nausea and stomach pain, which might lead to appetite changes and weight loss.
- Seizures, strokes, mental confusion, and brain injury are all risks.
- Respiratory disease.
- Memory, attention to detail, and decision-making challenges make daily living more challenging.
- One of the most devastating health consequences of drug abuse is death. Deaths from synthetic opioids and heroin have increased the most. In the previous year, 212,000 persons aged 12 and up had first consumed heroin. Every day, nearly 90 Americans are killed by an opioid overdose [21].
- Adverse effects of addiction on student brains [1, 20]:
 - Cocaine, nicotine, alcohol, marijuana, and other types of drugs have an influence on the brain's "reward" circuit, which is part of the limbic system. This area of the brain has an impact on instinct and emotion. Drugs disrupt this system by flooding the brain with enormous amounts of dopamine, a brain chemical that assists in the control of emotions and pleasurable feelings. This dopamine spike is what causes the "high." It is one of the leading causes of drug addiction.
 - Drug use may appear to be voluntary at first, and it can alter brain chemistry. This can change the way the brain works and damage a person's ability to make decisions. It has the potential to induce intense cravings and compulsive drug use. Over time, this tendency might lead to substance dependency or drug and alcohol addiction.
 - Alcohol affects the brain in both the short and long terms, as well as disrupting communication networks. These can influence mood, behavior, and other cognitive functions.
 - Alcohol-induced dietary deficiencies, seizures, and liver disease can all result in brain damage. According to research, alcohol-related brain problems may frequently be treated with appropriate therapy.
- Adverse effects of addiction on student behaviors:

Addiction may seriously affect a student's conduct, affecting both their personal and academic lives. Addicts at school, whether they be from drugs, alcohol, or other substances frequently suffer from poor judgment, impulsivity, and lack of self-control. Because of their inability to focus, greater procrastination, and less desire for academic tasks, this results in inferior academic accomplishment. Their mental health is also affected by addiction, leading to problems such as anxiety, sadness, and disturbed social relationships. Aggression and poor decision-making are other traits that might worsen academic decline and cause social difficulties.

Table 2 explains in detail the behavioral addiction theories of how drug use escalates and leads to various abnormal behaviors [1, 29]:

TABLE 1: Comparative table of related works.

Paper	Dataset	Methodology	Key limitations
[11]	Custom action dataset	Local space-time features + SVM	Limited to simple actions
[12]	Custom dataset	RGB + optical flow Streams + spatial overlap and softmax probability values	High computational burden
[14]	Human skeletal sequences	LSTM-based network	Lacks deep feature learning
[15]	UMN, UCSD-PED1	Transfer learning + SVM	Dataset specificity
[13]	J-HMDB-21	Single-shot multibox detector (SSD)	Limited real-time prediction
[18]	Dangerous behavior dataset	DangerDet + key points-based postures	Dataset and time-cost limitations
[16]	Custom surveillance dataset	CNN + RNN for abnormal behavior detection	Limited generalization, focused on isolated cases
[17]	Violent-flow dataset and web dataset	OCP + KNN	High computational complexity with KNN, limited to crowd dynamics
[19]	UCF-crime dataset	DenseNet121 + CNN	High computational resources required

TABLE 2: Abnormal behavior description related to addicted people.

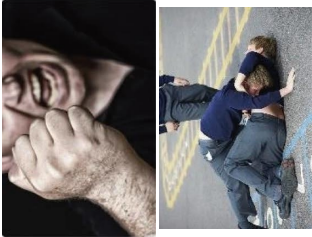
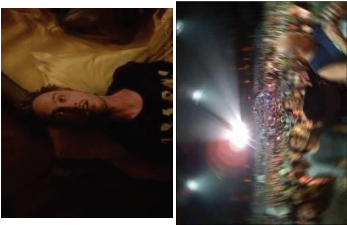
Behavior	Public image
- Paranoia: This is a mental disorder characterized by distrust and excessive suspicion of others, even when there is no reason to be suspicious [22]. In addition, Paranoia is a common mental health condition among drug users, particularly those who use stimulants such as cocaine and methamphetamine, has been shown in studies to severely disrupt interpersonal relationships and contribute to social isolation [23]	
- Aggressiveness: This is a tendency to real or fantasized conduct, with the objective of attacking or harming a living being or any object that hinders immediate satisfaction [24]. Moreover, drug-induced aggressiveness has been widely observed in people who abuse alcohol and stimulants, often resulting in physical altercations and legal consequences, and contributing to the individual's social and academic decline [25]	
- Hallucinations: These are inaccurate perceptions about objects or events that involve the senses. They seem true to the person experiencing them, but they cannot be confirmed by anyone else. Many types of hallucinogens: LSD¹, mescaline, psilocybin, PCP, cannabis, ecstasy, ketamine, salvia, and many more. The term "hallucinogens" refers to different drugs, also referred to as "psychedelic" drugs [1]	

TABLE 2: Continued.

Behavior	Public image
- Addictions: These cover a very broad area and include both behavioral addictions and substance addictions. Drug and alcohol abuse leads to both types of addiction. Behavioral addictions are characterized by the impossibility of controlling the practice of an activity. A feeling of increasing tension is set up before taking action, and at the time of the practice, the person feels a pleasure or a relief [22]	
- Impaired Judgment: Intoxication impairs brain function and judgment and can also alter mood. Depending on the drug, the person may experience a feeling of excessive excitement or well-being (euphoria) or may feel calmer, more relaxed, and more drowsy than usual [26, 27]	

TABLE 2: Continued.

Behavior	Public image
- Impulsiveness: These are the difficulties that a person has in managing to control themselves. Thus, impulsiveness regularly leads to violent acts or anger. These behaviors can in turn lead to undesirable consequences for their perpetrator [27, 28]	
- Loss of Self-Control: In a way, it is the addictive behavior that becomes “psychotropic” in itself. This change in behavior gradually leads to a loss of control, most often with repercussions on physical and mental health, as well as in the family, social, and professional domains	

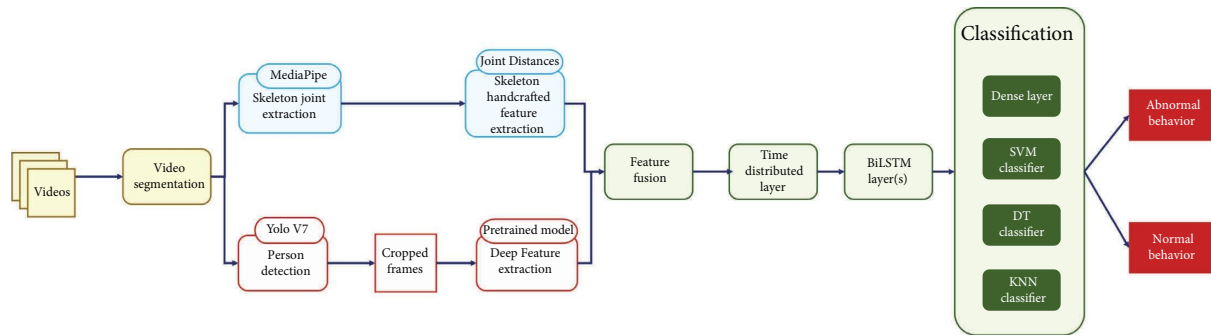


FIGURE 1: Evaluation framework architecture.

All these dangerous effects can risk the lives of adolescents, specifically in the schools and universities. For the body and brain effects, drug misuse can have serious physiological consequences, such as weakened immune systems, cardiac issues, seizures, strokes, lung disorders, and brain injury. Students suffering from addiction frequently struggle with memory, concentration, and decision-making, which impairs their academic and social functioning. These physiological deficiencies appear as visible behaviors in school settings. Examples include aggression and irritability, cognitive decline, physical symptoms, risk-taking, and criminal behavior.

To successfully detect and evaluate abnormal behaviors in drug-addicted students, we created a complete framework that makes use of powerful deep learning algorithms and computer vision models. In the next section, we will describe the methodological approach, which includes data preparation, feature extraction, and model design, to achieve high accuracy in behavior classification inside school contexts.

4. Methodology

The proposed evaluation framework, as well as the steps of the proposed study technique, is depicted in Figure 1. These steps are as follows:

- **Video Segmentation:** The input videos are first divided into individual frames using a video segmentation algorithm. This guarantees the temporal continuity of student activity while allowing for the analysis of each frame separately.
- **Handcrafted MediaPipe API Feature Extraction:** Using the MediaPipe API, we are able to extract information on each frame's skeletal joints, including 33 important skeletal joints like the elbows, knees, and shoulders. These joints provide the basis for handcrafted information such as joint locations (66 features, X and Y coordinates), Euclidean distances between subsequent joints (33 features), and joint movement velocity across frames (33 features). These handcrafted features offer precise spatial and temporal data on bodily movement, which is essential for recognizing abnormal behaviors. In addition, these features focusing on specific movements that correlate with stress, anxiety, and aggression. These movements are crucial indicators of abnormal

behavior in addicted students, and the feature extraction process ensures accurate modeling of these signs.

- **Deep Feature Extraction using Pretrained Models:** In parallel, we used pretrained deep learning models (YOLOv7 for body identification, MobileNetV2, ResNet100, Xception, VGG19, and InceptionResNetV2 for deep feature extraction). These models were used to extract high-level features from cropped frames, capturing abstract patterns of both normal and abnormal student behavior. These deep features complement the handcrafted features by providing a more detailed knowledge of behavior, especially in complex scenarios, and enhance the model's ability to distinguish between normal and abnormal behaviors.
- **Feature Fusion:** We then integrated the handcrafted features (skeletal joint information) with the deep features obtained by the pretrained models. This fusion enables us to capture both domain-specific and abstract behavioral representations, increasing the model's resilience in detecting abnormal behaviors.
- **Behavioral Classification Model:** The fused features are fed via a deep learning architecture that includes two TimeDistributed layers, one BiLSTM layer (for temporal dependencies), and numerous Dense layers with ReLu activation function and Dense layer with sigmoid activation function for classification purposes. This design guarantees that both spatial and temporal patterns in the student's behavior are accurately captured and evaluated as normal (Class: 0) or abnormal (Class: 1). In this step, we used cross-validation technique with $k = 5$, in which the data were divided into fivefolds, with each fold being used once for validation and the remaining four serving as training data. This technique was used to ensure that each fold maintains the original dataset's distribution of the target classes, addressing class imbalance concerns.

In addition, in this paper, the performance of our proposed methodology is evaluated by the well-known performance quantitative metrics (confusion matrix, accuracy, precision, recall, and F1-score) and qualitative indicators (class activation mapping [CAM]) through a comprehensive experimental study. In the following subsection, we describe in detail each step.



FIGURE 2: Samples from “Real-Life Violence Situations Dataset.” (a) Violence sample. (b) Nonviolence sample.

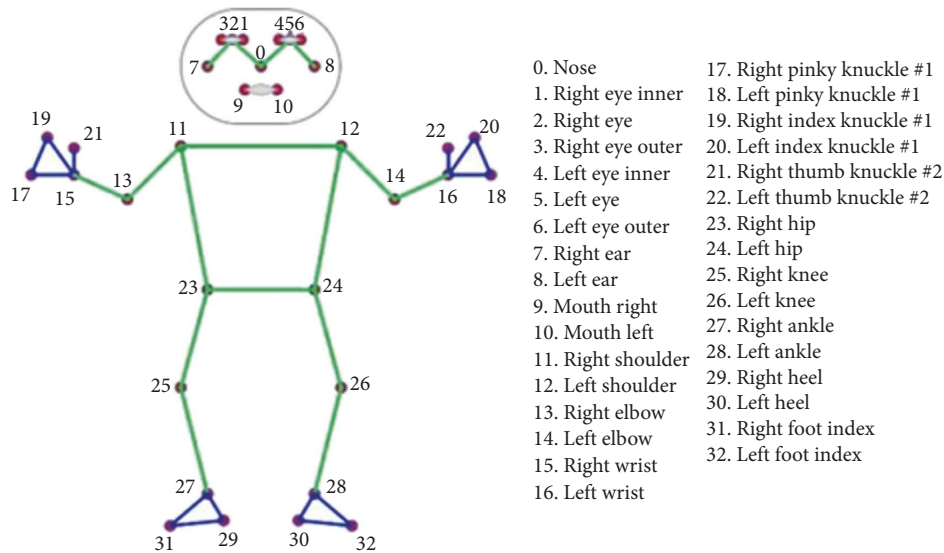


FIGURE 3: Thirty-three joints provided with the MediaPipe API³.

4.1. Real-Life Violence Situations Dataset Description and Preparation. It is important to note that implementing HAR based on computer vision techniques requires labeled datasets for training and evaluation. These datasets typically involve videos or images annotated with activity labels, which can be collected in controlled environments or real-world scenarios.

The video acquisition for addicted students in the school environment is a critical task seeing that various ethical concerns must be addressed due to the sensitivity of recognizing deviant behaviors in schools such as Privacy and Anonymity [30], Informed Consent [31], Data Protection and Security [32], Minimizing Bias and Misuse [8], and Transparency and Accountability [33]. So, given the lack of datasets for the addicted students in the school environment, we have resorted to use the “Real-Life Violence Situations Dataset.”³ Our choice for this dataset is justified by the fact

that it provides a wide variety of videos that show both violent and nonviolent behaviors, all of which are essential for correctly identifying addiction symptoms in an educational environment. In contrast to other datasets such as the UMN and UCF-Crime, which concentrate on various forms of conduct outside of schools, this dataset highlights actual instances of aggressiveness, particularly in educational settings. With this specific emphasis, our deep learning model becomes much more effective at identifying the subtle behavioral patterns associated with teenage drug usage, which makes it a useful tool for early identification and intervention.

Furthermore, we verified that all data acquisition methods followed stringent privacy and consent rules. Videos obtained from public platforms were anonymized to safeguard individuals’ identities, and no real-life student video was utilized in this study.

The “Real-Life Violence Situations Dataset”² contains 1000 movies of violence and 1000 videos of nonviolence acquired from YouTube. The violence videos in our dataset feature several severe violence situations in a variety of locales and conditions include various types of violent behaviors, such as physical altercations and aggressive gestures in a variety of settings, including the street, prison, and school, which are essential for training our model to detect similar behaviors in school environments. Nonviolence videos from our collection are also collected from a variety of human activities such as football, basketball, tennis, swimming, and eating. Each video was carefully annotated to reflect the presence or absence of violence, allowing for a structured approach to model training and evaluation. The gathered movies are deemed high resolution (480–720p) and cover a diverse range of people in terms of color, age, and gender, as well as distinct situations. Figure 2 displays some snippets from the produced dataset.

In our work, we must first perform the data pre-processing stage including many processes to assure data quality and improve model performance. Initially, the videos were divided into frames, which were then processed using the YOLOv7 model to detect and crop regions of interest, emphasizing the students’ bodies. Following that, we used the MediaPipe API to extract custom data such as joint distances, velocities, and position variations across frames. Their use influenced the decision to employ variables such as joint distances and joint velocities in identifying physical indications of stress, anxiety, and aggressiveness, which are prominent signs of drug addiction. These qualities enable the model to capture both spatial and temporal elements of behavior, resulting in a full study of student behaviors. The feature extraction method was carried out systematically, beginning with handcrafted features obtained from skeletal data and progressing to deep feature extraction with pre-trained CNN models. These attributes were then combined to employ both domain-specific and learned representations, enhancing the model’s capacity to discern between normal and abnormal behaviors.

After that, a data cleaning procedure was used to remove noise and irregularities, such as columns with significant percentages of missing or zero values, assuring the dataset’s robustness and reliability.

4.2. Handcrafted Feature Extraction Process. In this section, we will present in detail the second step of our proposed methodology. In this step, we employed the MediaPipe API in order to identify the skeleton joints. Then, set of handcrafted features is calculated.

4.2.1. MediaPipe API. MediaPipe is a well-known open-source framework for developing real-time multimodal machine learning applications such as computer vision, audio, and natural language processing [34]. One of

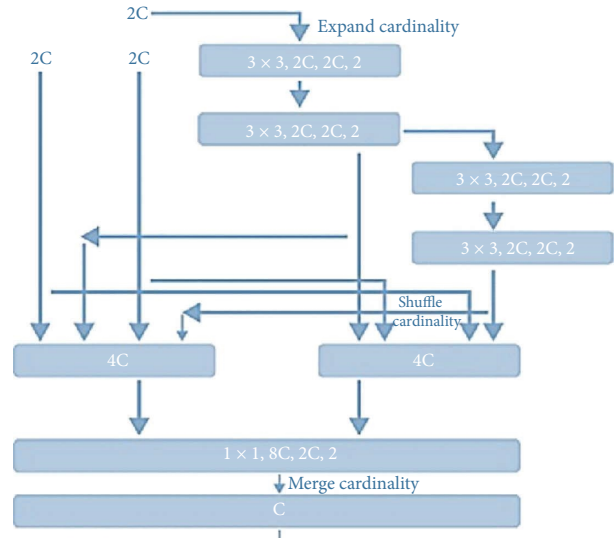


FIGURE 4: Extended efficient layer aggregation network (E-ELAN)⁴.

MediaPipe’s features is the ability to extract skeleton joint information from 2D and 3D photos and video frames. To extract skeleton joint information using MediaPipe, the developer can use the Pose Detection API, which provides a pretrained machine learning model that can detect skeleton joints on human bodies in real-time video streams. The Pose Detection API uses a deep neural network architecture to identify 33 skeleton joints, including the nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles (cf. Figure 3).

4.2.2. Skeleton Joint-Based Feature Extraction. MediaPipe posture is a high-fidelity body posture tracking system that uses RGB picture frames to create thirty-three 3D landmarks and a background segmentation mask over the entire body. It makes use of the BlazePose topology [35], which is a super dataset of the COCO [36], BlazeFace [35], and BlazePalm [37] topologies.

In this paper, once the skeleton joint information is extracted with the MediaPipe API, a set of handcrafted features is calculated as follows:

- Sixty-six skeleton joint positions: The provided joint is in the form of a set of (x, y) coordinates, representing the positions of each joint in the frame. A skeleton is formed by 33 joints. As a result, the Feature vector has 66 dimensions, where “X” is the location of the horizontal axis and “Y” is the vertical axis position.
- Thirty-three distances between two successive joints: These features are calculated based on the Euclidian distance between the pair of joints j and $(j + 1)$. These distances are calculated using the following formula:

$$\text{Distance between successive joints} = \sqrt{(X_{i+1} - X_i)^2 + (Y_{i+1} - Y_i)^2}. \quad (1)$$

TABLE 3: Pretrained deep learning models' description.

Model	Depth	Layers number	Image input size	Parameters (millions)
VGG19	19	47	224-by-224	144
ResNet101	101	347	224-by-224	44.6
InceptionResnetV2	164	825	299-by-299	55.9
Xception	71	171	299-by-299	22.9
MobileNetV2	53	155	224-by-224	3.5

- Thirty-three differences in distances between two subsequent frames: Using formula (2), the different

distances between two successive frames are calculated:

$$\text{Difference between successive frames} = \text{Distance}_{\text{current}} - \text{Distance}_{\text{previous}}. \quad (2)$$

- Thirty-three joint velocities: Considering the joint positions in two subsequent frames p_t^j and p_{t+1}^j , where “j” is the index of the joint position and “t” represents the timestamp of each frame of a video. Equation (3) is used to calculate the velocity relative to each joint [38]:

$$\text{Joint Velocity}_t^V = 1000 * \left\| p_t^j - p_{t+1}^j \right\|_2^2. \quad (3)$$

These handcrafted features provide detailed spatial and temporal information about body movement, which is critical for identifying abnormal behaviors such as aggression or lethargy. Furthermore, the “Real-Life Violence Situations Dataset” defined several scenarios for a person. For instance, an addicted student may fall on the floor and exhibit aggressive behaviors such as boxing fights toward students, teachers, and others. In addition, these students can commit robberies or crimes if they need money to get drugs and satisfy their cravings.

4.3. Deep Feature Extraction Process. In this section, we will present in detail the third step of our proposed methodology. In this step, we employed the YOLO V7 algorithm in order to detect people on videos. Then, we obtained the cropped frames based on the bounding box of the detected person. After that, we used the well-known pretrained models such as VGG19, Xception, ResNet100 and InceptionResNetV2, and MobileNetV2 in order to extract deep features.

4.3.1. YOLO V7 Model for Person Detection. YOLO is an acronym that stands for “You Only Look Once” and refers to a prominent family of real-time object detection methods. The first YOLO object detector was introduced in 2016 by Joseph Redmon et al. [39]. This architecture was substantially faster than the previous object detectors when it was released, and it quickly became the standard for real-time computer vision applications.

Since then, other YOLO versions and modifications have been presented, each with considerable improvements in performance and efficiency from YOLOv1 to finally YOLO v7. So, YOLOv7 is the most recent official YOLO version developed by the architects of the YOLO architecture [19]. We anticipate that many commercial networks will go

immediately from YOLO V4 to V7, skipping all other numbers. The YOLOv7 architecture [40] is based on the previous ELAN model architecture which is known as an extended efficient layer aggregation network (E-ELAN). ELAN enhances model learning ability by employing cardinality to expand, shuffle, and merge without disturbing gradient flow paths. In terms of architecture, it simply affects the architecture in the computational block, while the architecture in the transition layer stays the same as ELAN (cf. Figure 4). E-ELAN employs group convolution to increase the computing block's channels and cardinality. The same channel multiplier and group parameter is applied to all computational blocks in a computing layer. Each computing block's feature map will be randomized into groups of g and then concatenated together. Finally, to accomplish merge cardinality, a shuffled group feature map will be introduced.

In our work, we employed the YOLOv7 for its excellent real-time object identification capabilities, which are critical for precisely recognizing and cropping frames of student activities in video sequences. Then, all the obtained bounding boxes are cropped and considered as cropped frames as input to the pretrained model to extract features. This step is detailed in the next subsection.

4.3.2. Deep Feature Extraction. CNNs have been shown to be a strong tool for image classification and feature extraction in the field of deep learning. Similarly, a pretrained CNN model can be used for feature extraction from images that can be used for several application fields such as image similarity, object detection, people detection, and image retrieval.

In this paper, we resort to employing the most popular pretrained models from each family like VGG19, Xception, ResNet101, InceptionResNetV2, and MobileNetV2 (see Table 3).

The VGG19 [41] is a deep CNN model with 19 layers that employ small convolution filters (3×3). Increasing the network depth from 16 to 19 layers greatly increases performance over earlier designs. Xception [42] is built on depthwise separable convolutions, which decouple cross-channel and spatial correlations, resulting in better performance than the Inception design. VGG19 and Xception

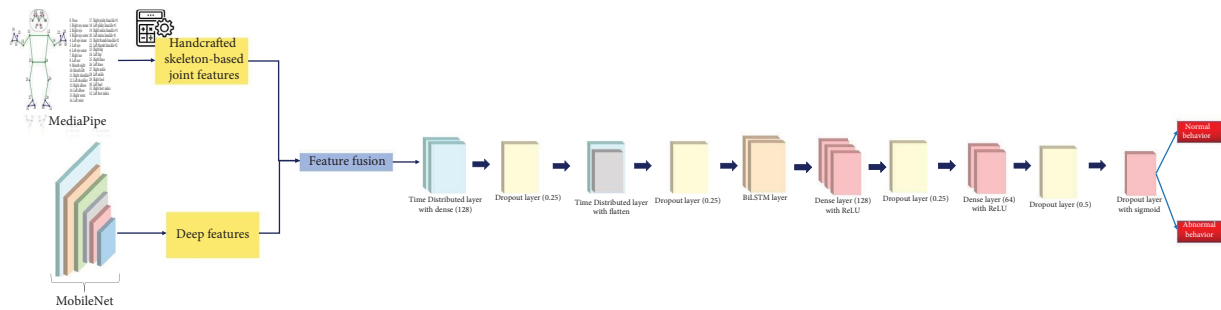


FIGURE 5: Deep Network architecture to classify fused features.

were chosen for their architectures' resilience in feature extraction, which is especially effective in discriminating between normal and abnormal behaviors.

ResNet101 [43, 44] uses residual learning and skip connections to train deep networks. It employs different strides in the convolution layers than ResNet50, resulting in higher accuracy and a modest performance cost. InceptionResNetV2 [45] combines Inception V3 with ResNet, utilizing residual connections to ease deep network training. ResNet101 and InceptionResNetV2 were chosen for its depth and capacity to learn complicated features, which are essential for detecting subtle and complex behavioral inputs.

MobileNetV2 [46] is tailored for mobile applications, with an inverted residual structure and depthwise convolutions for increased efficiency. It increases mobile model performance and serves as the foundation for lightweight mobile semantic segmentation models such as Mobile DeepLabv3. MobileNetV2 was selected for its efficiency and ability to function effectively on mobile and embedded devices, which is important given the system's practical implementation in real-world contexts.

In this paper, we propose using the transfer learning technique, which is the reuse of pretrained networks, which involves employing the weights of pretrained networks' first layers to train deeper layers on the "Real-Life Violence Situations Dataset." Because these pretrained networks know how to identify lower level features such as edges, lines, and curves. Along with convolutional layers, which are often the most time-consuming elements of the computing process, adding these weights allows the network to achieve an exceptional degree much faster than beginning from scratch.

4.4. Feature Fusion. Feature fusion involving handcrafted features and deep features refers to combining information extracted from traditional, manually engineered features (handcrafted) with features learned automatically by deep neural networks (deep features). This approach aims to leverage the strengths of both types of features to improve the performance of a machine learning model or system. Several methods can be used to combine these different types of features as follows:

- Concatenation or Stacking: Simply concatenating or stacking the handcrafted and deep features to create a joint feature representation.

- Weighted Fusion: Assigning different weights to handcrafted and deep features based on their importance or relevance, then combining them linearly or nonlinearly.
- Feature Level Fusion: Applying operations (such as element-wise addition, subtraction, multiplication, or division) to combine corresponding elements of two feature sets.
- Late Fusion: Feeding handcrafted features and deep features separately into different parts of the model and combining their outputs at a later stage in the network.

Moreover, the feature fusion method offers several advantages as follows: (1) giving complementary information seen that handcrafted features may capture domain-specific knowledge, while deep learning learns abstract representations and (2) improved performance by combining both types of features which can enhance the robustness and generalization of the model, potentially leading to better performance. However, there are many challenges that arise when applying the fusion feature methods like (1) feature compatibility with ensuring that the features from different sources are compatible and can be effectively fused and (2) dimensionality and overfitting with combining diverse features might increase the dimensionality of the data, requiring careful handling to avoid overfitting.

In this paper, in order to capitalize on the strengths of both types of feature representations, we simply adopt the concatenation method to fuse the extracted handcrafted and deep features extracted with MobilNetV2 pretrained model. Specifically, the concatenation method was chosen for its simplicity and efficiency in fusing features from both sources, ensuring compatibility without additional complexity. This fusion method is critical to increasing the performance of the suggested model by merging the capabilities of both handcrafted and deep feature sets, allowing for a more robust analysis of the data. The handcrafted features are particularly useful for domain-specific behavior analysis, which correlates precise physical movements with behavioral cues. Deep features allow the model to understand complex visual cues, which are critical for distinguishing between normal and abnormal behaviors in a variety of contexts. So, the concatenation fusion method takes advantage of the strengths of both feature sets handcrafted for domain-specific spatiotemporal details and deep features for abstract, high-level representations to

emphasize how the model gains a more holistic understanding of behavior, resulting in increased accuracy and robustness in abnormal behavior detection.

After that, we preprocessed our dataset by eliminating feature columns with empty values. So, we set a threshold for the percentage of zeros in a dataset, calculate the percentage of zeros in each column, and then filter out columns that have a high percentage of zeros or missing values. Then, we removed these filtered columns from the *DataFrame*. Finally, these features are considered the input of the proposed deep network which will be described in the next subsection.

4.5. Deep Model Architecture. A deep model architecture refers to the structure and arrangement of neural network layers that comprise a deep learning model (cf. Figure 5). The choice of layers, including TimeDistributed, BiLSTM, and multiple Dense layers, was based on empirical studies from prior experiments in HAR and behavioral analysis in computer vision as follows [47, 48].

- **Input and Feature Representation:**

The model takes a sequence of features as input, expressed as concatenated feature vectors. These vectors, obtained from a variety of sources, including deep and handcrafted features, are structured with dimensions of (None, 128, 284,332). This input format represents the temporal nature of the data since 128 frames are processed sequentially.

- **TimeDistributed Layer:**

To handle this input format, Keras offers a layer known as TimeDistributed (Dense (128, (input_shape))) with Dense that wraps a Dense layer applied to each time step in the sequence. Then, a Dropout (0.25) layer is added for regularization purposes to prevent overfitting. This choice was inspired by previous research showing that dropout significantly improves model robustness in deep learning architectures, particularly in tasks involving behavior classification. After that, another TimeDistributed (Flatten ()) is incorporated into the model to flatten the output to prepare the next layers.

- **Bidirectional Long Short-Term Memory (BiLSTM) Layer:**

We used an LSTM layer to capture temporal aspect between frames. This layer analyzes sequential data to identify patterns that emerge over time. So, we employed a *BiLSTM* to process input in both forward and backward directions, improving the model's capacity to capture dependencies from past and future frames. Following the LSTM layer, a Dropout layer (0.25) is added for additional regularization.

- **Multiple Dense Layers:**

After extracting temporal features using the LSTM, a multiple Dense layers are added (256, 128, 64, and 32 units) with *ReLU* activation function followed by

Dropout layer for regularization. These layers likely serve as intermediary representations, capturing hierarchical information from sequences.

- **Final Classification:**

Finally, a Dense layer with a sigmoid activation function for binary classification (0: normal behavior and 1: abnormal behavior) given the use of "binary_crossentropy" loss function.

- **Training and Optimization:**

The compilation of our model is performed by using Adam optimizer [49] with a learning_rate=0.0005, beta_1=0.9, beta_2=0.999, and epsilon=1e-08. These values were chosen based on a combination of theoretical considerations and empirical tuning to achieve optimal performance. Initially, we followed standard values for Adam's hyperparameters as suggested in the literature, given their widespread use and proven effectiveness for deep learning tasks.

Additionally, to fine-tune the learning rate, we conducted a grid search over multiple values (ranging from 0.0001 to 0.001) and observed that a learning rate of 0.0005 resulted in the best trade-off between training speed and convergence stability. The beta values (0.9 and 0.999) and epsilon (1e-08) were selected based on their default settings, as these values have been shown to work well in similar contexts, particularly for optimizing deep neural networks with adaptive moment estimation, as referenced in [37]. To ensure robustness, we validated the model performance using a fivefold cross-validation process, which confirmed the suitability of these hyperparameters for our specific dataset and model architecture. Furthermore, we considered epochs=20 and batch_size=8 as the other fine-tuning hyperparameters.

5. Experimental Study

In this section, an assessment was conducted to choose the most effective model for detecting abnormal behaviors in addicted students. The evaluation focused on spatiotemporal handcrafted features, spatial deep features, and the fusion of spatiotemporal features (both handcrafted and deep). The following set of experiments will be elaborated upon to detail the outcomes.

5.1. Experiments With Spatiotemporal Handcrafted Features.

We performed two sets of experiments: experiments with spatial feature extraction and other experiments with spatiotemporal feature extraction. In this experiment, we extracted and evaluated a set of 15 continuous spatial features, five face properties provided by Kinect and 10 Euclidean distances.

In this experiment, we extracted and evaluated a set of 99 continuous spatiotemporal handcrafted features (33 Euclidean distances between two successive joints; 33 difference distances between two subsequent frames; and 33 joint velocities) (see Section 4.2.2). These features are considered

TABLE 4: Classifier's configurations and hyperparameters' details.

Classifier	Hyperparameters	Optimization technique	Best configuration
K-nearest neighbors (KNN)	- Number of neighbors (k): 3, 5, 7, 9 - Distance metric: Euclidean	5-Fold cross-validation	$k = 5$, Euclidean distance
Random forest (RF)	- Number of trees: 50, 100, 200 - Maximum depth: 10, 15, 20 - Criterion: Gini	Grid search + 5-fold cross-validation	100 trees, max depth = 15, Gini criterion
Support vector machine (SVM)	- Kernel: Linear, polynomial, RBF - Regularization (C): 0.1, 1, 10 - Gamma (RBF): 0.1, 0.01, 0.001	Grid search + 5-fold cross-validation	RBF kernel, $C = 1$, gamma = 0.01
Naïve Bayes (NB)	- No hyperparameters (Gaussian NB used)	No tuning required	Gaussian Naïve Bayes

TABLE 5: Classification results.

Experiment	Features	Feature extraction method	Model	Classifier	Accuracy (%)
1	Handcrafted features	MediaPipe		Naïve Bayes	56.6
2	Handcrafted features	MediaPipe		RF	65.5
3	Handcrafted features	MediaPipe		SVM	89.7
4	Handcrafted features	MediaPipe		KNN	76.5
5	Deep feature	MobileNetV2	MobileNetV2 + LSTM	Dense layer with sigmoid	93.33
6	Deep feature	InceptionResNetV2	InceptionResNetV2 + LSTM	Dense layer with sigmoid	81.21
7	Deep feature	VGG19	VGG19 + LSTM	Dense layer with sigmoid	84.14
8	Deep feature	Xception	Xception + LSTM	Dense layer with sigmoid	80.13
9	Deep feature	ResNet100	ResNet100 + LSTM	Dense layer with sigmoid	79.56
10	Spatiotemporal fused features	Handcrafted features + features extracted with MobileNetV2	TimeDistributed + BiLSTM + Dense layers	Dense layer with sigmoid	97.5

Note: The bold value represents the best result.

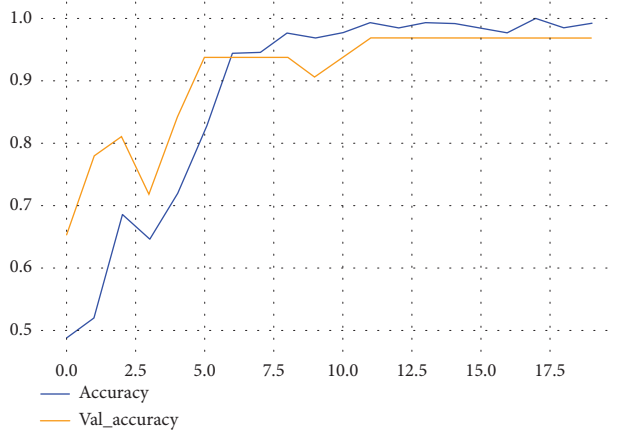


FIGURE 6: Training_Accuracy and Validation_Accuracy results.



FIGURE 7: Training_Loss and Validation_Loss results.

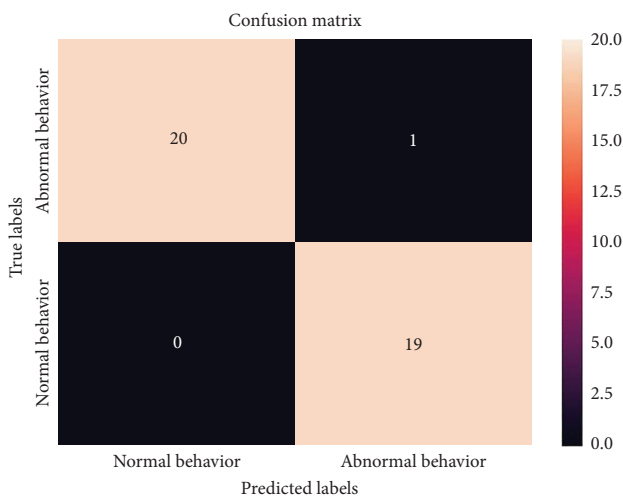


FIGURE 8: Confusion matrix of normal and abnormal behavior.

the input of machine learning classifiers such as KNN, RF, SVM, and NB. Table 4 details the classifiers, their hyper-parameters, and the optimization techniques. The experimental study recorded a significant and acceptable difference that varies from 56.6% with the NB classifier to 89.7% with the SVM classifier (see Table 5).

5.2. Experiments With Deep Features. In order to evaluate the “Real-Life Violence Situations Dataset,” we employed the well-known baseline CNN models on our dataset such as Resnet100, VGG19, Xception, InceptionResnetV2, and MobileNetV2 to extract deep features from the cropped frames segmented with YOLOV7, which we considered only the person class to detect person in frames. Then, we used the cropped frames as the inputs of the well-known pretrained models. After that, the extracted features are considered the input a TimeDistributed, LSTM, and Dense layers. According to Table 5, the overall recorded results fall within a range of 79.56% with ResNet100% and 93.33% with MobileNetV2.

5.3. Experiments With Fused Handcrafted and Deep Features. In order to take advantage of the deep features and the meaning of the proposed (handcrafted) features, and given that fusion affects the robustness and performance of the proposed model, we resort to fusing the two types of features (handcrafted and deep). Seen that the extracted deep features with the MobileNetV2 pretrained model give us the best results for classification, we proposed to fuse these features with the handcrafted features and consider them as the input of a model composed of 2 TimeDistributed Layers, 1 BiLSTM Layer, and several Dense layers, as detailed in Section 4.4. This model gives us efficient results with an accuracy of 97.5%.

To understand how well our proposed model is performing during training and validation, we resort to displaying the accuracy and the loss curves. As seen in Figures 5 and 6, X-axis represents the epochs and the Y-axis represents the Accuracy and Loss.

In Figure 6, we note that throughout the training phase, the accuracy curve exhibits a constant upward trend, indicating that the model’s ability to categorize normal and abnormal actions properly increases with time. This increasing movement indicates that the model is learning successfully and converging to better accuracy. As the curve approaches a plateau, it suggests that the model has achieved its peak performance, with no further substantial gain in accuracy over subsequent epochs.

In Figure 7, similarly, the loss curve consistently decreases, indicating that the model lowers its prediction errors. This drop in the loss function demonstrates that the model’s predictions are improving with time. The continuous decline in loss without unpredictable spikes demonstrates that the learning process is stable, with no serious difficulties such as divergence or instability detected.

TABLE 6: Quantitative evaluation results.

Class	Accuracy	Precision	Recall	F1-score
0: Normal behavior	0.975	1.00	0.95	0.98
1: Abnormal behavior	0.975	0.95	1.00	0.97

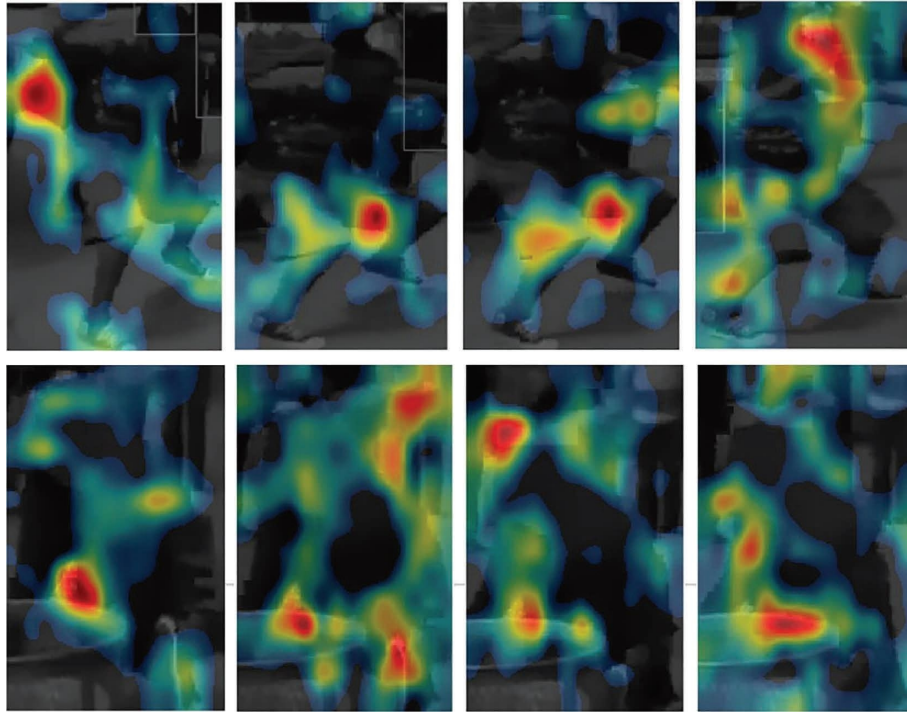


FIGURE 9: Region contributions of MobileNetV2 predictions using CAM.

Our model presents a good generalization by comparing the training and validation curves for accuracy and loss. We can see that the training and validation curves follow similar trends. So, we can conclude that our model is free of overfitting and underfitting.

In conclusion, the smooth progression of both the accuracy and loss curves and the alignment between the training and validation sets point to the robustness of the proposed model. The model demonstrates stable learning behavior, achieving its best performance at an accuracy of 97.5% without any anomalies.

The suggested model is scalable because it employs efficient transfer learning models such as MobileNetV2, making it suited for large-scale applications with limited computing resources, such as real-time student behavior monitoring. By integrating deep features with computationally efficient handcrafted features, the technique can handle bigger datasets without experiencing substantial delays. We understand the value of computing resources, and thus, our system uses pretrained models such as MobileNetV2 and ResNet101, which are designed for modest GPU hardware. Testing using midrange GPUs, such as the NVIDIA RTX 2050, revealed that real-time apps can process frames at roughly 20 milliseconds. While the proposed model shows promise in controlled situations,

variables such as student behavior, lighting, and camera angles may influence performance in real-world scenarios. To solve these issues, we want to include new data modalities and fine-tune the system for certain circumstances.

6. Quantitative and Qualitative Evaluation

To assess our proposed approach, we used well-known performance quantitative metrics and qualitative indicators through a comprehensive experimental study. In the next subsections, we will highlight the quantitative and qualitative evaluation metrics and results used in our work.

6.1. Quantitative Evaluation. To evaluate our abnormal behavior for addicted students, approach it, and compare it with the previous related works, we used the following well-known performance/evaluation metrics.

- **Confusion matrix:** A confusion matrix, known as a contingency table, is a tool for measuring the performance of a machine learning model and its predictive accuracy compared to real classification problems. Figure 8 represents the confusion metrics performed by comparing the real classification of normal and abnormal behavior with the predictive classification.

TABLE 7: Comparative study with the state of the art.

Paper	Feature types	Dataset	Model	Accuracy
[15]	Spatial deep features	- University of Minnesota (UMN) dataset - University of California dataset - San Diego Pedstrain1 (UCSD-PED1) dataset	VGGNet-19 + BSVM	- 97.95% for (UMN) dataset - 86.69% for University of California dataset - 79.87% for UCSD-PED1 dataset
[16]	Spatiotemporal deep features	Film videos	CNN + RNN	95.3%
[17]	Spatiotemporal handcrafted features	- Violent-flow dataset - Web dataset	KNN classifier	- 98.90% for violent-flow dataset - 99.62% for web dataset
[18]	Handcrafted features	Dataset from surveillance films	DangerDet, a unique end-to-end dangerous behavior detection technique	71.0%
[19]	Deep features	UCF-crime dataset	DenseNet121 + CNN	86.63%
Our	Handcrafted + deep features	Real-Life Violence Situations Dataset	MobileNet + BiLSTM + Dense layers	97.5%

- Accuracy rate [50]: This is the correct classification of classified videos as normal behavior or abnormal behavior. It allows for the calculation of the overall classification performance.
- Precision [50]: Precision, also called Confidence, denotes the proportion of predicted positive cases that are correctly real positives.
- Recall [50]: It is referred to as the true-positive rate or sensitivity. It is defined as the ratio of the total number of correctly classified positive/abnormal behavior instances divided by the total number of positive/abnormal behavior instances.
- F1-score [50]: It is the harmonic mean of precision and recall.

Table 6 shows that for the normal behavior class, the model had an accuracy of 97.5%, a perfect precision score of 1.00, a recall of 0.95, and an F1-score of 0.98. These findings show that the model has a strong capacity to properly identify normal behavior instances, with a small margin of error in recall, where 5% of normal examples were misclassified. The model also achieved 97.5% accuracy in the abnormal behavior class, with a precision of 0.95, suggesting that 5% of the occurrences identified as abnormal were incorrectly categorized. However, the recall for this class was 1.00, indicating that all aberrant cases had been appropriately detected. The F1-score for the abnormal behavior class was 0.97, indicating the balance of accuracy and recall. These detailed results confirm the robustness of the model in distinguishing between normal and abnormal behaviors, demonstrating strong, balanced performance across all evaluation metrics.

6.2. Qualitative Evaluation. CAM, introduced by authors in [51], is a method that utilizes visual representations to forecast CNN's decisions. In our study, we used CAM to visually emphasize the regions of the input image that were most important to the model's decision-making process. The CAM technique generates a heatmap representing the class of interest. We retrieved the feature maps from the final convolutional layer and used global average pooling to obtain class-specific weights. These weights were then used to calculate a weighted sum of the feature maps, resulting in a class activation heatmap that identifies the discriminative areas for the anticipated class. In Figure 9, we analyze the predictions of the MobileNetV2 network using CAM. In this analysis, MobileNetV2 accurately identifies the image class. The class activation mapping visuals demonstrate the involvement of different regions within the input image concerning the predicted class "abnormal behavior." The red regions indicate the most influential areas. Notably, the network's prediction relies on the identification of interest in skeleton joints, distances, and velocities, as proposed in the handcrafted method.

The heatmaps' correspondence with quantitative measures demonstrates how certain regions, such as skeletal joints and their movements, match the model's classification results. For example, the model identified abnormal behaviors with 97.5% accuracy, as seen in the heatmaps, which reveal strong activity in regions associated with drug-

induced behaviors such as anger and anxiety. These regions of interest directly contribute to the model's excellent accuracy and recall scores, with heatmaps concentrating on joints related to aggressive gestures showing strong recall when identifying such behaviors. Furthermore, further analyses strengthen the model's effectiveness by illustrating how essential traits, as represented in heatmaps, correspond to quantitative measurements, including case examples in which the model properly diagnosed abnormal behaviors. The model's robust performance, as evidenced by an F1-score of 0.97, is supported by the constant localization of these behavioral indicators.

The comparative insights with other state-of-the-art methods further emphasize how the integration of handcrafted and deep features not only improves classification accuracy but also results in heatmaps that effectively represent underlying behaviors, particularly those related to addiction. This comparative study is detailed in the following section.

7. Comparative Study With the State of the Art

To achieve a more realistic evaluation, we set side by side our suggested approach not only a few existing works of abnormal behavior for addicted students in school but also other recent works that focused on treating abnormal behaviors of people. According to Table 7, our suggested strategy, which uses a fusion technique to combine handcrafted and deep features, obtained 97.5% accuracy on the "Real-Life Violence Situations Dataset." This performance is comparable to other state-of-the-art approaches, but with clear advantages. For example, the authors in [15] achieved 97.95% accuracy on the University of Minnesota (UMN) dataset with VGGNet-19 and a Binary SVM, but its performance drops significantly to 86.69% for the University of California dataset and 79.87% for the UCSD-PED1 dataset, indicating that its generalizability across different datasets is limited when compared to our method. Similarly, [16] employed spatiotemporal deep features (CNN + RNN) and got 95.3% accuracy on film video. While the use of spatiotemporal characteristics results in robust temporal modeling, this method does not include handcrafted features. In contrast, our strategy makes use of both handcrafted and deep features, which improves the model's capacity to capture both domain-specific nuances and abstract representations, resulting in a more thorough detection of abnormal behaviors. According to [17], the KNN classifier obtained 98.90% accuracy on the violent-flow dataset and 99.62% on the web dataset, both of which rely on spatiotemporal constructed features. While these findings are outstanding, relying solely on handcrafted features may restrict adaptation to more complicated settings in which deep features might give stronger discriminating. Our combination of the two feature types finds a compromise between their strengths, resulting in greater resilience in real-world settings. In contrast, approaches such as [18] (reaching 71.0% accuracy using handcrafted features on surveillance videos) and [19] (achieving 86.63% accuracy using deep features on the UCF-Crime dataset)

show limits in terms of performance or dataset specificity. While these approaches are useful in some circumstances, they do not take advantage of the combined strength of handmade and deep features, as our methodology does, resulting in more competitive accuracy across varied scenarios. Despite our suggested approach's benefits, we must note its computational complexity as a result of the combination of deep and handcrafted features. This might be interpreted as a disadvantage when compared to models such as [16, 19], which are quicker but achieve significantly lower accuracy. However, our method's improved accuracy and greater application justify the computational cost trade-off.

8. Conclusions and Future Works

In this study, we introduced a novel method aimed at identifying abnormal behaviors among addicted students in school settings. Our approach focused on developing a predictive model to create an intelligent detector for such behaviors, addressing an increasingly crucial need in educational environments. This system not only improves the identification of addiction-related behaviors but also prepares the path for future integration into educational systems, which might provide a preventive approach to drug misuse. Aside from the technological contributions, this study emphasizes the necessity of early detection techniques in ensuring students' academic and personal growth. The suggested approach, by notifying educational professionals and guardians, might play an important role in reducing the spread of addiction and its accompanying consequences.

Our empirical validation, using the "Real-Life Violence Situations Dataset," confirmed the system's ability to recognize abnormal behavior. Handcrafted features from skeletal joint recognition using the MediaPipe API were combined with classifiers including KNN, RF, SVM, and NB. The accuracy of these classifiers ranged from 56.6% for NB to 89.7% for SVM, demonstrating the system's ability to successfully use spatial features for behavior recognition.

In addition to handcrafted features, we used transfer learning models to extract deep features from CNN architectures such as VGG19, Xception, ResNet100, and InceptionResnetV2. These features were processed using TimeDistributed, LSTM, and Dense layers, resulting in accuracy ranges of 79.56% to 93.33%, with MobileNetV2 surpassing the others. The combination of handcrafted and deep features resulted in a final model architecture with an accuracy of 97.5%, demonstrating the strength of integrating spatial and temporal data for abnormal behavior classification.

While the current study establishes a solid framework for recognizing abnormal student behavior, some limitations must be addressed. First, the model is primarily reliant on publicly accessible dataset, such as the "Real-Life Violence Situations Dataset," which may not adequately reflect the range of behaviors unique to school environments. As a result, future research will focus on creating a specific dataset suited to addicted students in real-world educational environments. This dataset will comprise not just behavioral

actions, but also multimodal inputs such as facial expressions, voice analysis, and physiological indications such as heart rate. Furthermore, environmental circumstances such as camera positioning, illumination, and resolution can all have an influence on the model's present accuracy, affecting real-time detection performance. Future enhancements will focus on fine-tuning the system so that it can perform well in a variety of educational settings, including low-resource contexts. Finally, more sophisticated strategies, such as model assembly and the incorporation of other data modalities, will be investigated to improve the model's resilience and generalizability. We also want to include user input from educational professionals to continually increase the system's sensitivity to specific behaviors associated with addiction and other mental health concerns.

Data Availability Statement

Real-Life Violence Situations Dataset is available in the kaggle website for research purposes (<https://www.kaggle.com/datasets/mohamedmustafa/real-life-violence-situations-dataset>).

Disclosure

The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Conflicts of Interest

The authors declare no conflicts of interest.

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Endnotes

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